

A Step-Selection Trilemma for Finite-Difference Generator Curvature

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Abstract

Second-order generator geometry is in live use: geodesic interpolation, curvilinear edits, and Hessian disentanglement priors all reach for the curvature of the generated manifold, and the estimator a practitioner reaches for is a central finite difference. We show that on real generators this estimator faces a step-selection *trilemma*: the steps that best preserve curvature magnitude, signed bias, and the ranking of directions are systematically different, so no single step gets all three right at once. The three criteria are optimized at three different steps (magnitude-best $\delta = 3.0$, bias-best $\delta = 2.0$, rank-best $\delta \approx 0.2$). The magnitude error has a $\sqrt{\varepsilon_{\text{mach}}}$ floor (classical; Proposition 1 proves it exists), measured at $\geq 45\%$ on StyleGAN (47% on a matched fp32/fp64 control, so precision does not remove it); the signed bias crosses zero between the magnitude- and rank-best steps; and a fixed-step finite difference recovers the ranking well (Spearman up to 0.98). The conflict reproduces across three StyleGAN-bedroom seeds, three full-resolution FFHQ seeds, and lower-detail controls, at generator-dependent step values. As an exact reference we use composed forward-mode automatic differentiation – standard nested-JVP machinery (Pearlmutter, 1994; Fike & Alonso, 2011; Baydin et al., 2018; Cobb et al., 2024), applied to generator curvature by Lee & Park (2023) – step-free and exact to machine precision (validated to 2.5×10^{-17} against an analytic ground truth, vs the best finite difference’s 8.6×10^{-9}). Using this reference we run a replication audit of whether curvature predicts edit nonlinearity beyond first-order displacement; the signal is significant on two of five anchors but does not transfer (cross-anchor 95% CI $[-0.09, +0.43]$ spans zero), and a fixed-step finite difference returns the same non-transferring answer. The instrument’s value is on the magnitude and sign axes, where no step is correct, rather than on this rank-level question. We recommend that second-order generator-geometry reports use an exact or step-validated estimator for magnitude and sign, report the step axis, and replicate direction-level claims across anchors.

1 Introduction

Editing methods for pretrained GANs increasingly treat the latent space as a Riemannian manifold and reach for its second-order structure. Geodesic interpolation replaces linear interpolation using a pullback metric (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018); curvilinear edit paths are constructed to be nonlinear by design (Aoshima & Matsubara, 2023); and second-order Hessian information is used as a disentanglement prior (Peebles et al., 2020). Each of these needs the curvature of the generator – the change of the Jacobian along an edit – and the estimator a practitioner reaches for is a finite-difference second difference of the generator.

That estimator has no defensible setting. A curvature estimate must get three things right – its magnitude, its sign, and the ranking of directions it induces – and for the second-order difference on StyleGAN these three are optimized at *different* steps: no single step preserves all three at once, and the signed error even changes sign across the usable range (Section 4, Table 2). The first-order difference of the same code has no such conflict; one step region is at once accurate, unbiased, and rank-faithful, which is why first-order Jacobians and pullback metrics have been safe to finite-difference while second-order curvature is not. The magnitude axis of this conflict is an irreducible relative-error floor of order $\sqrt{\varepsilon_{\text{mach}}}$ (Proposition 1): on

StyleGAN no shared step beats a 45% mean error (38% median), identical in fp32 and fp64, so the source is the generator’s nonlinearity rather than roundoff. Because no step is defensibly choosable, finite differences cannot yield a curvature one can trust for magnitude, sign, and ranking at once. The same qualitative split repeats on two additional StyleGAN-bedroom seeds and on three full-resolution StyleGAN-FFHQ seeds, with lower-detail FFHQ controls (Appendix D); the particular best steps change, but the conflict between magnitude, bias, and ranking does not.

The fix is exact. The second-order term is the directional derivative of the first, so composing forward-mode automatic differentiation with itself returns the second-order pushforward to machine precision – no step, in memory a small constant multiple of one forward pass (Section 5). We validate it against a first-order finite-difference control that agrees to 2.4% and against an analytic referee where it reaches 2.5×10^{-17} while the best finite difference manages only 8.6×10^{-9} .

With the exact second-order quantity in hand, free of the step-selection confound, we revisit a concrete question a practitioner faces – does the curvature of an edit direction predict how *nonlinear* the edit is, how far the rendered path bows away from the straight chord? – as an audit of what the instrument changes.

Our main finding is an empirical step-selection trilemma for finite-difference generator curvature; we establish it against an exact composed-JVP reference and use that reference to audit one curvature–nonlinearity claim. Concretely:

- An empirical step-selection *trilemma* for finite-difference generator curvature (the paper’s main contribution): magnitude, signed bias, and rank fidelity are optimized at different steps (Table 2), while the first difference of the same code is accurate, unbiased, and rank-faithful together near one step (2.4%). Its magnitude axis is the classical $\sqrt{\varepsilon_{\text{mach}}}$ second-difference floor (Higham, 2002; Press et al., 2007), restated as Proposition 1 and measured at $\geq 45\%$ on StyleGAN (identical in fp32/fp64); the bias and rank axes are empirical. The conflict repeats on two more bedroom seeds and three full-resolution FFHQ seeds with lower-detail controls (Sections 3, 4; Appendix D).
- An exact reference for that measurement: composed forward-mode AD – standard nested-JVP machinery (Pearlmutter, 1994; Fike & Alonso, 2011; Baydin et al., 2018; Cobb et al., 2024), used by Lee & Park (2023) for generator curvature – which is step-free and exact to machine precision at a small constant multiple of one forward pass (Section 5).
- A deterministic prescription: probing second-order generator geometry calls for the exact instrument; any method built on a finite-difference generator Hessian – the Hessian Penalty (Peebles et al., 2020) – inherits the per-sample step-selection ambiguity (Section 7).
- A scope delineation (Section 6): the floor is a property of the estimator, and it bites where curvature *magnitude* or *sign* enters a computation directly, not where only the cross-direction ranking matters. We make this concrete on one ML application – predicting edit nonlinearity from curvature – where the exact reference and a fixed-step finite difference return the same (non-transferring) answer, confirming that on a rank-level task the floor does not change the conclusion.

The estimator gap therefore matters for magnitude- and sign-dependent uses of generator curvature (geodesic energy, curvature-penalty weights); a step-validated finite difference or the exact reference is needed there, while rank-level uses tolerate finite differences.

2 Setup

Let $G : \mathbb{R}^d \rightarrow \mathbb{R}^{3HW}$ be a pretrained generator and z a latent code. For a unit direction v the edit path is $\alpha \mapsto G(z + \alpha v)$. We compare three quantities along this path at $\alpha = 0$.

Prediction target (estimator-free). The *path nonlinearity* $NL(z, v)$ is how far the rendered path departs from the straight chord between its endpoints, measured by rendering only: sample K points $\alpha_k \in [-T, T]$, form the chord between $G(z - Tv)$ and $G(z + Tv)$, and average the normalized image-space deviation of the

interior points from the chord. This is exactly “how wrong is linear interpolation” over the edit range, and it uses no derivative estimator, so it cannot favor either proxy.

First-order proxy (baseline). The displacement $\|J_z v\| = \|\partial_\alpha G\|$, the exact first-order JVP. This is the cheap quantity a practitioner already has.

Second-order proxy. The curvature magnitude $\|\partial_\alpha^2 G\|$. Whether this predicts NL beyond the first-order proxy is the audit we return to in Section 6; obtaining it reliably is the problem the instrument solves.

3 Second differences cannot recover generator curvature

The obvious way to obtain $\partial_\alpha^2 G$ without automatic differentiation is a central second difference, whose accuracy is governed by the classical truncation-versus-cancellation tradeoff (Higham, 2002; Press et al., 2007). It does not work in the precision generators run in.

Proposition 1 (Second-order non-recoverability). *Let $f \in C^4$ near α with $f^{(4)}$ bounded and $f''(\alpha) \neq 0$, evaluated in floating point with unit roundoff $\varepsilon_{\text{mach}}$ under a standard rounding model (Higham, 2002). The central second difference $D_\varepsilon^2 f = (f(\alpha + \varepsilon) - 2f(\alpha) + f(\alpha - \varepsilon))/\varepsilon^2$ has relative error*

$$\frac{|D_\varepsilon^2 f - f''(\alpha)|}{|f''(\alpha)|} \lesssim C_T \varepsilon^2 + C_R \varepsilon_{\text{mach}}/\varepsilon^2, \quad C_T = \frac{|f^{(4)}(\alpha)|}{12|f''(\alpha)|}, \quad C_R = O\left(\frac{|f(\alpha)|}{|f''(\alpha)|}\right),$$

whose minimum over the step is of order $\sqrt{C_T C_R \varepsilon_{\text{mach}}}$, attained at $\varepsilon^* \sim \varepsilon_{\text{mach}}^{1/4}$; no step does better. The central first difference, by contrast, attains a floor of order $\varepsilon_{\text{mach}}^{2/3}$. For a vector map $G : \mathbb{R}^d \rightarrow \mathbb{R}^{3HW}$ the bound holds componentwise, so a single global step is governed by the worst component.

The truncation term $\frac{\varepsilon^2}{12} f^{(4)}$ and the cancellation term $O(\varepsilon_{\text{mach}} |f|/\varepsilon^2)$ cannot be small together; the exponent $\frac{1}{2}$ is intrinsic to the operation and the constant carries the generator’s nonlinearity $|f^{(4)} f|/f''^2$, which is large for a deep network (derivation in Appendix A). In double precision the floor is about 10^{-8} in the benign case; in the single precision generators use ($\varepsilon_{\text{mach}} \approx 1.2 \times 10^{-7}$) it is about 3×10^{-4} benign and far worse for a strongly nonlinear G , where Section 4 measures $\geq 45\%$. The first difference is different: a milder floor $\varepsilon_{\text{mach}}^{2/3}$ with a wide usable window, so first-order Jacobians are safe to finite-difference while second-order curvature is not. This asymmetry is also our control: wherever the first-order finite difference agrees with the exact JVP, the pipeline is correct and any second-order disagreement is real.

To check the constants against ground truth, let $f(\alpha) = [e^{0.7\alpha}, \sin 3\alpha, (\alpha + 0.4)^3, \tanh 2\alpha, \alpha^2 \cos \alpha]$, with derivatives known in closed form. At $\alpha = 0.3$ in double precision (Table 1) the central second-order relative error is U-shaped, bottoming at 8.6×10^{-9} near $\varepsilon = 10^{-4}$ and diverging to 0.29 by $\varepsilon = 10^{-8}$; the composed JVP reaches 2.5×10^{-17} , so the best finite difference is 3.5×10^8 times worse, matching the Proposition 1 bound with the 1/12 constant.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 1: Second-order accuracy on the analytic toy at $\alpha = 0.3$, double precision. No finite-difference step approaches the exact composed JVP.

step δ	role	second order (FD vs exact)			first order	
		mag. err	signed bias	Spearman	mag. err	Spearman
8.0	large step	74%	−74%	0.71	64%	0.69
3.0	2nd mag-best	45%	−26%	0.90	44%	0.87
2.0	2nd bias-best	56%	+5%	0.93	37%	0.90
0.2	2nd rank-best	355%	+350%	0.983	9%	0.995
0.02	1st-order stable	6773%	+6773%	0.72	2.4%	0.999

Table 2: Step-selection trilemma for second-order generator curvature on StyleGAN-bedroom (96 direction–anchor pairs; central finite difference vs the exact composed-JVP). The steps that best preserve magnitude ($\delta = 3.0$), signed bias ($\delta = 2.0$), and rank fidelity ($\delta = 0.2$) are different, and the signed bias crosses from underestimation at large steps to overestimation at small steps. The analogous first-order quantity (right columns) is accurate, unbiased, and rank-faithful in a common region near $\delta = 0.02$. Bold marks the per-criterion optimum. Full 16-step sweep in Appendix C.

4 The non-recoverability is real on a generator

On StyleGAN-bedroom at 256×256 in fp32, the best single step $\delta \approx 3$ shared across edit directions leaves the central second-difference curvature with a 45% mean relative-error floor (38% median across directions), rising on both sides and exploding as $\delta \rightarrow 0$ (Figure 1). On matched inputs the floor is identical in fp32 and fp64 (47% in both; the gap from the 45% best-step value reflects the input sample, not precision), confirming it is truncation against the generator’s nonlinearity rather than roundoff. This $\geq 45\%$ figure is the *magnitude* axis of Proposition 1; on a real generator the difficulty is three-fold.

An empirical step-selection trilemma. Magnitude accuracy is not the only thing a second-order estimate must get right. A practitioner may need the curvature *magnitude* (geodesic energy, regularizer strength), its *sign*/unbiasedness (a thresholded decision), or the *ranking* of directions by curvature (which edit is “most curved”). Across our 96 direction–anchor pairs these three criteria are optimized at *different* steps (Table 2): magnitude error is least at $\delta = 3.0$ (45%), the signed bias is smallest at $\delta = 2.0$ (+5%), and rank fidelity is highest at $\delta \approx 0.2$ (Spearman 0.983). The magnitude-optimal step gives lower rank fidelity than the rank-optimal step, while the rank-optimal step incurs a 355% magnitude error; and the signed bias crosses from underestimation at large steps to overestimation at small steps, so there is no “safe side” to err on. This three-way conflict is *empirical* – Proposition 1 bounds only the magnitude axis, and the bias and rank axes are measured rather than proved – and it is specific to second order: the analogous first-order quantity is accurate, unbiased, and rank-faithful together in a common region near $\delta = 0.02$ (2.4% error, −0.0% bias, Spearman 0.999; right columns of Table 2). The first-order central difference of the same code thus agrees with the exact JVP to 2.4%, so the pipeline is correct and the second-order conflict is genuine. The exact composed-JVP instrument removes the step axis altogether and serves as the reference against which all three criteria are even defined. These specific step values are properties of this generator’s nonlinearity; what we expect to generalize is the *existence* of the conflict – a magnitude floor coexisting with an opposite-signed cancellation regime – not the particular steps, which is why we state the trilemma as empirical.

Robustness. Repeating the same 16-step sweep on two additional bedroom seeds preserves the conflict: the magnitude-best step remains $\delta = 3.0$ with 51% and 43% mean error, while the rank-best steps move to $\delta = 0.2$ and $\delta = 0.3$ and incur 315% and 258% magnitude error. On full-resolution StyleGAN-FFHQ, three seeds repeat the pattern: the magnitude-best step is $\delta = 5.0$ with 71%, 71%, and 69% mean error; the signed-bias optimum is $\delta = 1.5$; and the rank-best steps move to $\delta = 0.3$, 0.1, and 0.2, incurring 266%, 763%, and 358% magnitude error. Lower-detail FFHQ controls at lod 1 and 2 preserve the qualitative split, and a separate exact-JVP FFHQ resolution audit preserves the attribute ranking of second/first-order curvature ratios across lod 0, 1, and 2 (all pairwise Spearman correlations are 1.0). Thus the robustness claim is qualitative: the trilemma repeats, not the exact step values (Appendix D).

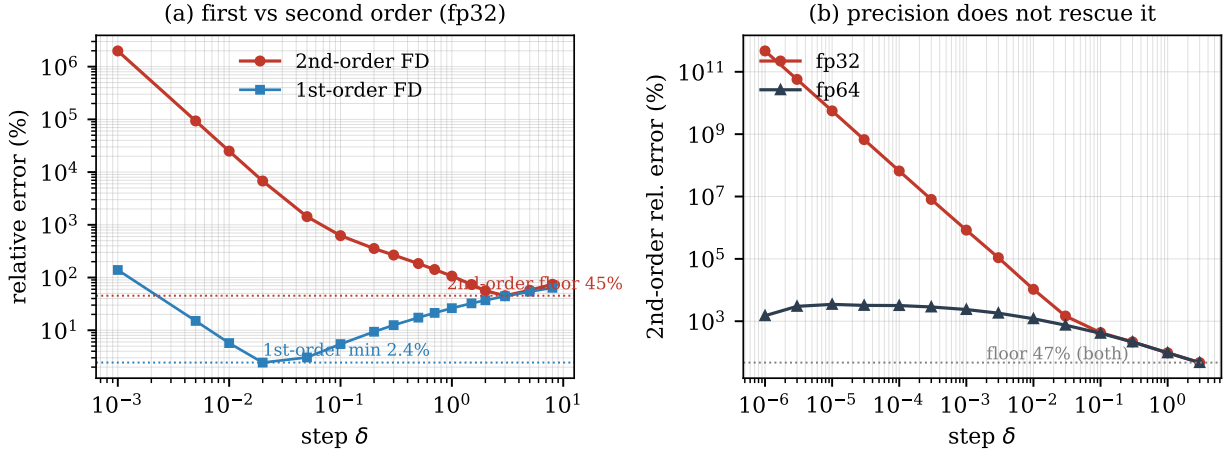


Figure 1: Non-recoverability of Proposition 1 on StyleGAN. The first-order finite difference (blue) has a usable minimum; the second-order difference (red, mean over directions) never falls below $\approx 45\%$ at any common step, and the floor is identical in fp32 and fp64.

5 An exact, step-free instrument

The second-order term is the directional derivative of the first, so composing forward-mode AD with itself,

$$\partial_\alpha^2 G(z + \alpha v) = \partial_\alpha [\partial G(z + \alpha v)[v]][v],$$

is the exact derivative of the implemented graph and is evaluated by nesting two JVP calls (in PyTorch, `torch.func.jvp` of a function that returns a JVP) (Paszke et al., 2019; Griewank & Walther, 2008). At generator scale, the required object is the per-direction second-order pushforward of the implemented network. Composed/forward-mode AD gives that object directly, building on established ingredients – Hessian-vector products at gradient cost (Pearlmutter, 1994), hyper-dual numbers for exact second derivatives without a step (Fike & Alonso, 2011), and second-order forward-mode AD (Baydin et al., 2018; Cobb et al., 2024) – and we use it here as the exact reference against which finite differences are audited. No Jacobian or Hessian is materialized, so memory is a small constant multiple of a single forward evaluation – about two for the second-order term – rather than scaling with the output dimension, and k nested sweeps give the k -th-order pushforward. The instrument needs only a differentiable forward graph and applies to any such architecture (a short per-generator adapter sufficed in our experiments). Its purpose here is to make the prediction question of Section 6 answerable with the *exact* second-order quantity, so that whatever the answer turns out to be cannot be blamed on the estimator.

6 A replication audit the instrument makes answerable

We now use the exact instrument to ask a concrete question: does an edit direction’s curvature predict the estimator-free nonlinearity NL beyond the first-order displacement? This is exactly the kind of audit the finite-difference estimator’s step-selection ambiguity (Section 4) would make uninterpretable, since a changed answer could be an artifact of the estimator. The exact pushforward removes that ambiguity, so the answer is about the geometry. We use StyleGAN-bedroom, 50 edit directions (the 8 annotated attribute boundaries plus 42 random w -directions), and average each quantity over 12 latent anchors per direction; we repeat the entire correlation across five independent anchor seeds and report the distribution. Rank correlations are Spearman; partial correlations residualize ranks; confidence intervals are bootstrap over directions; the first-order control of Section 4 passes throughout (2.4%).

A single anchor and a hand-picked subset cannot decide the question. On one anchor set restricted to $N=24$ *named* directions – a subset of the fifty, and a different population from the full mixed set we use

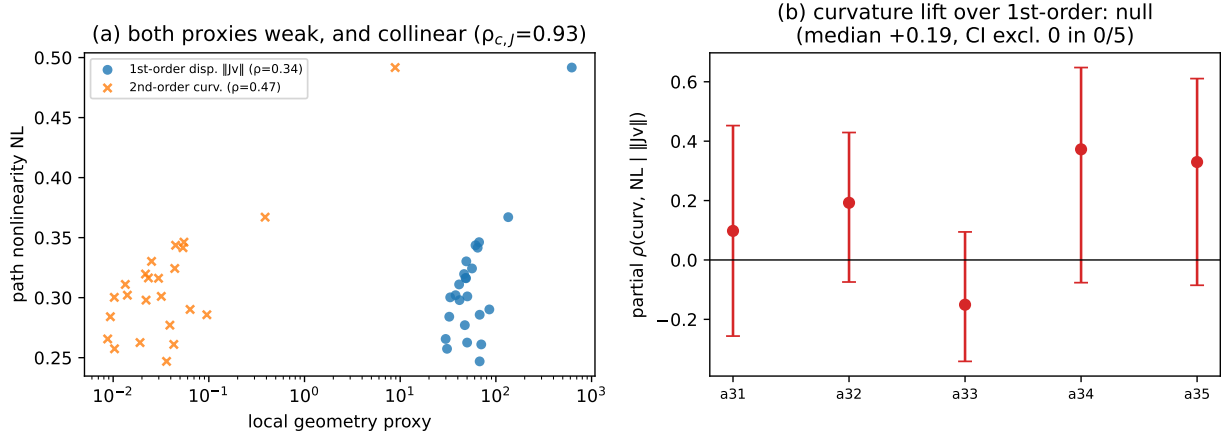


Figure 2: The exact second-order signal for edit nonlinearity is anchor-dependent and does not transfer. (a) On a single anchor, first-order displacement and second-order curvature are nearly collinear predictors of path nonlinearity. (b) Across five anchors, the partial correlation of curvature given displacement is significant on 2/5 by permutation (seeds 34, 35), while each anchor’s bootstrap 95% CI includes zero; the cross-anchor estimate +0.17 has a CI that spans zero, so the effect does not transfer.

across anchors below – exact curvature correlates with nonlinearity at $\rho = 0.47$ ($p = 0.012$), with partial correlation +0.45 ($p = 0.016$, 10^4 permutations). This is not a robust signal: displacement and curvature are nearly collinear ($\rho_{c,J} = 0.93$), two heavy-tailed directions carry it (removing them drops the raw correlation to 0.31, $p = 0.075$), and it is computed on a hand-picked subset. We report it only as the forking-paths trap it is – a strong-looking number that the proper analysis below dissolves.

Across anchors the signal does not transfer. On five anchors (50 directions each, Figure 2b) the partial correlation $\rho(\text{curv, NL} \mid \|J_z v\|)$ is +0.10, +0.19, −0.15, +0.37, +0.33. We report both tests symmetrically. By permutation the per-anchor p -values are 0.25, 0.095, 0.86, 0.0038, 0.010, so the signal is significant on two of five anchors, and both survivors clear a Bonferroni threshold of .01. By bootstrap, on the other hand, every anchor’s 95% CI includes zero (e.g. seed 34 [−0.076, +0.648], seed 35 [−0.085, +0.611]). The two tests do not agree – permutation flags two anchors, every bootstrap CI includes zero – and with only $n = 5$ anchors and CIs this wide the data are equally consistent with a small positive effect and with none; the honest reading is that the signal does not transfer. The cross-anchor mean partial is +0.17 (unweighted mean of the five per-anchor estimates; 95% CI from a Student- t on $n = 5$), [−0.09, +0.43], falling to +0.06 ([−0.23, +0.34], median +0.09) once chord length is also controlled. So the effect does not transfer reliably to a new anchor.

Reading the audit. The exact second-order signal is significant on two of five anchors but does not transfer: the second-order statistic should not be treated as anchor-stable without replication. The $\geq 45\%$ figure is a *magnitude* error, whereas these audit correlations are rank/residual statistics. The instrument-necessity is therefore the trilemma of Section 4: a fixed-step finite difference’s curvature cannot be trusted for scale, sign, and ranking at once, and the practitioner cannot know in advance which step preserves the property a given audit needs. A nuanced second-order effect is therefore reliably auditable with an exact or step-validated reference, which removes the step choice. We checked this confound directly: a fixed-step finite-difference curvature recovers the exact direction ranking (Spearman 0.82 at $\delta = 3.0$ rising to 0.96 at $\delta = 0.1$) and two of the three exact top directions at small steps (one of three at $\delta = 3.0$), and against the estimator-free nonlinearity it tracks the exact curvature ($\rho_{\text{FD}} = 0.19\text{--}0.35$ across steps vs $\rho_{\text{exact}} = 0.47$) rather than flipping the verdict. So for this *rank-level* audit the two estimators agree on the (non-transferring) answer; the exact reference’s necessity is on the magnitude and sign axes (Section 4, Table 2), where no step is correct, not on this ranking question.

Why the question is a priori reasonable. The result is not forced by collinearity: the chord length – raw global displacement – is uncorrelated with nonlinearity ($\rho = 0.065$), so NL is not merely a re-encoding of how far the edit moves. The leading non-trivial term of the chord deviation is $\frac{1}{2}\alpha^2\partial_\alpha^2 G$, so second-order curvature is the *a priori* right-order predictor of nonlinearity; the instrument lets us test that prediction honestly rather than through the estimator’s floor, and the honest answer is the mixed, non-transferring one above.

7 Recommendations, scope, and limitations

Recommendation 1: use the exact instrument for second-order generator geometry. Probing second-order generator geometry calls for the exact instrument. This is a deterministic consequence of Proposition 1: a finite-difference generator Hessian has a $\geq 45\%$ relative-error floor on StyleGAN that no step size or precision upgrade (including fp64) removes, while the composed-JVP pushforward is exact to machine precision at the cost of a small constant multiple of one forward pass. The same floor governs any method built on a finite-difference second derivative of a generator. The Hessian Penalty (Peebles et al., 2020) estimates exactly this second difference $(1/\varepsilon^2)[G(z+\varepsilon v) - 2G(z) + G(z-\varepsilon v)]$; it is a stochastic training-time regularizer, so averaging across samples may temper the error, but each per-sample estimate it averages sits on the floor of Proposition 1. For per-sample second-order measurement, the exact composed-JVP Hessian-vector product removes the step choice at a comparable constant number of generator evaluations.

Recommendation 2: report the step axis when finite differences are used. If a finite-difference generator Hessian is used anyway, one tuned step is not enough evidence. The report should show, or rule out, the three-way conflict we measure in Table 2: magnitude error, signed bias, and rank fidelity can be optimized at different steps. Without that audit, the step choice can decide whether a magnitude, sign, or ranking conclusion holds.

Recommendation 3: replicate direction-level second-order claims across anchors. The replication audit in Section 6 shows the second failure mode: single-anchor variability can decide whether a curvature correlation is significant. A second-order generator-geometry claim should therefore pair an exact or validated estimator with replication across anchors before treating a direction-level effect as stable.

On the audit. The second-order signal for predicting edit nonlinearity is real in aggregate but anchor-dependent (Section 6); it does not transfer reliably across anchors, so for a practitioner who cannot fix the anchor regime, first-order displacement $\|J_z v\|$ – a single JVP – is the safe default. That the signal is auditable at all is itself a result for the instrument: a fixed-step finite difference cannot be trusted for the scale, sign, and ranking of curvature at once (Section 4), so modest partial-correlation estimates could not be reliably attributed to geometry rather than step choice without the exact reference.

Scope and limitations. The prediction audit is confined to what we measured – StyleGAN-bedroom, the path-nonlinearity target, and curvature versus first-order displacement – and the cross-anchor effect is modest (+0.17, CI spanning zero), so it stands only within this setting; we did not run the nonlinearity-prediction study on FFHQ, diffusion U-Nets, or other targets such as identity preservation and attribute leakage. The finite-difference trilemma is more broadly audited: it repeats on two additional bedroom seeds, three full-resolution FFHQ seeds, and lower-detail FFHQ controls, but its exact step values and floor heights are generator-dependent. The non-recoverability result (Proposition 1) is architecture-independent and the instrument applies to any differentiable generator.

8 Related work

Latent-geometry analyses of generative models use the first-order pullback metric $G = J^\top J$ to define geodesics and argue that geodesic interpolation improves on linear interpolation (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Yang et al., 2018); these methods apply a curved path everywhere and rely on the generator’s higher-order structure without separating how reliably that structure can be

computed. Curvilinear editing (Aoshima & Matsubara, 2023) constructs nonlinear edit paths to be commutative. Second-order information appears as a training-time disentanglement prior in the Hessian Penalty (Peebles et al., 2020), which estimates the Hessian by exactly the finite-difference second difference our Proposition 1 shows is non-recoverable. Closest to our object, Lee & Park (2023) regularize the explicit (extrinsic) curvature of a deep generative decoder using second-order automatic differentiation, with a stochastic Hutchinson trace estimator on low-dimensional autoencoders as a training-time prior; we instead compute the exact per-direction second-order pushforward of pretrained production-scale generators as a measurement instrument and quantify when the finite-difference alternative fails. Recent work computes curvature of generative manifolds by careful grid finite differences (Koulakis & Seibold, 2026), a regime distinct from the single-anchor second difference studied here. Our contribution is to identify the empirical trilemma that this classical step-size analysis does not by itself settle on real generators: finite differences cannot preserve curvature magnitude, sign, and rank with one defensible step, while an exact composed-JVP reference makes the resulting second-order claims auditable. The numerical ingredients are established (Press et al., 2007; Higham, 2002; Pearlmutter, 1994; Fike & Alonso, 2011; Griewank & Walther, 2008; Baydin et al., 2018; Cobb et al., 2024); the new object is the generator-scale measurement decision they expose.

9 Conclusion

Finite differences cannot reliably recover the curvature of a generator. In single precision the central second difference has a $\geq 45\%$ magnitude-error floor that no step or precision removes (Proposition 1), and on a real generator this widens into an empirical step-selection trilemma: the steps that best preserve magnitude, signed bias, and rank fidelity are different, while the first-order quantity stays accurate, unbiased, and rank-faithful in a common step region (Section 4). The exact composed-JVP instrument recovers it, to machine precision and at the cost of a small constant multiple of one forward pass. The prescription follows directly: probing second-order generator geometry calls for the exact instrument, since any method built on a finite-difference generator Hessian depends on an unvalidated step choice. We then use the instrument to audit one practical question: a single-anchor curvature–nonlinearity finding that looked strong becomes an aggregate but non-transferring effect under anchor replication. That outcome is a decision rule for future second-order reports: use an exact or step-validated estimator first, then replicate direction-level claims across anchors before treating them as stable (Locatello et al., 2019; Hewitt & Liang, 2019).

Reproducibility statement

The submitted source archive contains the fixed-seed metrics behind every number in Sections 3–6, the pure re-aggregation script for Tables 2–3, and the FFHQ runner scripts that record the GPU protocols used for Table 4. The evidence directory contains the per-anchor prediction tables, the per-record finite-difference sweeps, and the robustness metrics and summaries. The Section 6 confound check (finite-difference versus exact ranking and decision recovery) is reproduced by `30_curvature_ordering_flip.py` and `31_curvature_decision_groundtruth.py` with their fixed-seed JSON outputs (`curvature_ordering_flip_metrics.json`, `curvature_decision_gt_metrics.json`). The main generator is the public StyleGAN-bedroom-256 checkpoint (Karras et al., 2019); the robustness sweep uses the public InterFaceGAN StyleGAN-FFHQ semantic boundaries and checkpoint at lod 0–2 (Shen et al., 2020). The generator runners require the public checkpoints and repository wrappers; the fixed JSON artifacts allow the submitted tables to be verified without rerunning GPU experiments. All experiments run on a single 8 GB GPU.

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A Derivation of Proposition 1

Taylor expansion of $f(\alpha \pm \varepsilon)$ to fourth order gives $D_\varepsilon^2 f = f''(\alpha) + \frac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)$ for the exact arithmetic, the truncation term. In floating point each evaluation carries error $O(\varepsilon_{\text{mach}}|f|)$; dividing the differenced numerator by ε^2 amplifies it to $O(\varepsilon_{\text{mach}}|f|/\varepsilon^2)$, the cancellation term. Summing and minimizing over ε gives $\varepsilon^* \sim (\varepsilon_{\text{mach}} C_R / C_T)^{1/4} \sim \varepsilon_{\text{mach}}^{1/4}$ and floor $\sim \sqrt{C_T C_R \varepsilon_{\text{mach}}}$. The first difference divides by ε , giving cancellation $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, optimum $\varepsilon_{\text{mach}}^{1/3}$, floor $\varepsilon_{\text{mach}}^{2/3}$.

B Experimental setup

Path nonlinearity uses $T = 3$ and $K = 33$ render points. The second-order proxy is the mean absolute $\partial_\alpha^2 G$ via two nested `torch.func.jvp` calls; the first-order proxy is the JVP norm. The finite-difference trilemma sweep in Table 2 uses three HiGAN attribute boundaries (indoor lighting, wood, view), three random unit w -directions, and 16 base latents (seed 31) (Yang et al., 2020). Its robustness sweeps repeat the same bedroom protocol on seeds 32 and 33 and run an analogous InterFaceGAN StyleGAN-FFHQ protocol (Shen et al., 2020) with five semantic boundaries, three random directions, and 12 base latents at full resolution (lod 0, seeds 31–33), plus lower-detail lod 1 and 2 controls (Appendix D). The multi-anchor prediction analysis uses the 8 HiGAN attribute boundaries (placed in their default manipulation layers) and 42 random unit w -directions, anchor seeds 31–35, and 12 base latents each; permutation tests use 10^4 resamples; bootstrap CIs use 2000 resamples over directions.

C Full finite-difference step sweep

Table 2 in the body shows five representative steps. Table 3 below gives the complete 16-step grid behind it, for both the second- and first-order finite differences against the exact composed-JVP, averaged over all 96 direction–anchor pairs; the body’s five rows are a subset, not a selection. Each criterion’s optimum (least magnitude error, least |signed bias|, highest Spearman) is reached at a different second-order step, while the first-order columns are jointly near-optimal across a broad region around $\delta = 0.02$ – 0.05 .

step δ	second order (FD vs exact)			first order (FD vs exact)		
	mag. err	signed bias	Spearman	mag. err	signed bias	Spearman
8.0	74.5%	−74.4%	0.712	63.8%	−63.8%	0.687
5.0	57.3%	−55.6%	0.818	54.0%	−54.0%	0.770
3.0	45.2%	−26.2%	0.897	44.1%	−44.1%	0.867
2.0	56.1%	+4.9%	0.934	37.1%	−37.1%	0.903
1.5	73.6%	+31.7%	0.950	32.4%	−32.4%	0.924
1.0	106.4%	+76.1%	0.965	26.2%	−26.2%	0.948
0.7	142.0%	+121.6%	0.975	21.3%	−21.3%	0.967
0.5	184.2%	+170.3%	0.981	17.4%	−17.4%	0.980
0.3	266.9%	+258.7%	0.983	12.5%	−12.5%	0.991
0.2	355.2%	+350.4%	0.983	9.4%	−9.4%	0.995
0.1	622.8%	+621.0%	0.970	5.5%	−5.5%	0.999
0.05	1,429%	+1,429%	0.906	3.1%	−2.9%	0.999
0.02	6,773%	+6,773%	0.721	2.4%	−0.0%	0.999
0.01	25,023%	+25,023%	0.575	5.7%	+4.5%	0.996
0.005	93,346%	+93,346%	0.527	15.0%	+14.4%	0.986
0.001	1,985,856%	+1,985,856%	0.409	139.1%	+139.0%	0.765

Table 3: Complete 16-step finite-difference sweep on StyleGAN-bedroom (96 direction–anchor pairs), second- and first-order, against the exact composed-JVP. The second-order magnitude optimum ($\delta = 3.0$), signed-bias optimum ($\delta = 2.0$), and rank optimum ($\delta \approx 0.2$ – 0.3) are at distinct steps; the first-order quantity is jointly accurate, unbiased, and rank-faithful around $\delta = 0.02$ – 0.05 .

D Robustness sweeps for the trilemma

Table 4 repeats the trilemma summary across two additional bedroom seeds, three full-resolution StyleGAN-FFHQ seeds, and two lower-detail FFHQ controls. A separate FFHQ exact-JVP resolution audit preserves the ranking of semantic attributes across lod 0, 1, and 2 (Spearman 1.0 for each pair). The step values are generator-dependent. The repeated pattern is that the second-order magnitude, signed-bias, and rank optima separate, while the first-order control remains near a common small-step region.

sweep	n	2nd mag best	2nd bias best	2nd rank best	1st stable
bedroom seed 31	96	3.0/45%	2.0/ + 5%	0.2/355%	0.02/2.4%
bedroom seed 32	96	3.0/51%	1.5/ + 11%	0.2/315%	0.02/2.2%
bedroom seed 33	96	3.0/43%	2.0/ + 5%	0.3/258%	0.02/2.3%
FFHQ lod 0 seed 31	96	5.0/71%	1.5/ + 2%	0.3/266%	0.02/8.0%
FFHQ lod 0 seed 32	96	5.0/71%	1.5/ − 8%	0.1/763%	0.02/7.2%
FFHQ lod 0 seed 33	96	5.0/69%	1.5/ + 1%	0.2/358%	0.02/7.5%
FFHQ lod 1 seed 31	96	8.0/74%	2.0/ + 12%	0.3/360%	0.02/10.0%
FFHQ lod 2 seed 31	96	8.0/72%	3.0/ − 13%	0.3/340%	0.02/8.7%

Table 4: Robustness summary for the empirical step-selection trilemma. Each cell reports step/error, except the bias column, which reports step/signed bias. The FFHQ rows test whether the qualitative conflict survives a different generator and semantic-boundary set; the lod 1 and 2 rows serve as lower-detail controls.